

App Feed Algorithm Design Document

Social Media Platform — Feed, Navigation & Recommendation System

Version 1.0 | Product Design | 2025

Contents: Navigation Flow • Feed Algorithm • Video & Posts Algorithm

1. App Navigation Flow

This section describes the top-level routing of the application based on authentication state.

1.1 Authentication Routing

IF NOT LOGGED IN	IF LOGGED IN
→ Login Screen ↓ Home Screen	↓ Home Screen

1.2 Home Screen Navigation Tabs

P — Photos	V — Videos	M — Messages	U — User Profile
Photo Feed	Video Feed (Coming Soon)	Direct Messages (Coming Soon)	Profile & Settings

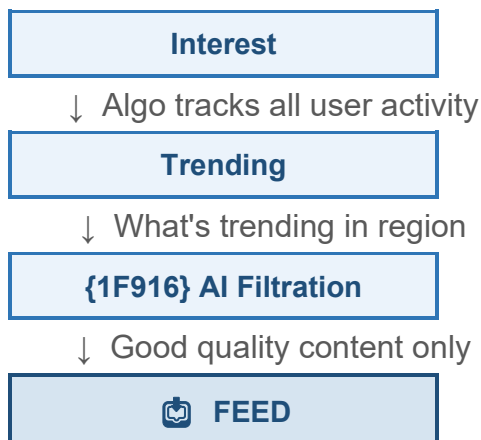
2. Feed Algorithm

The feed uses two parallel algorithms selected randomly each session. Access requires location, interest, and following permissions.

Access Permissions

- Location
- Interest ✓
- Following ✓

2.1 Algorithm ① — Interest-Based Feed



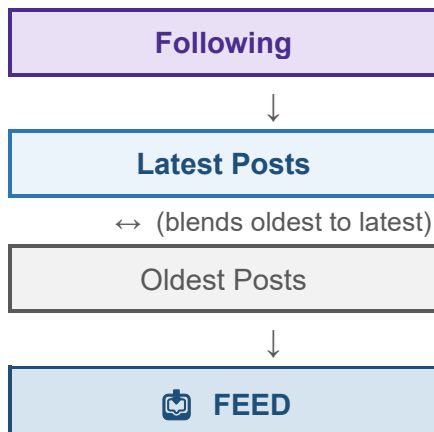
Stage Notes

Interest: Algorithm tracks ALL user activity and interests demonstrated over time.

Trending: Identifies what's currently trending in the user's region.

📱 AI Filtration: An AI model scores and filters content for quality, relevance, and safety before posts reach the feed.

2.2 Algorithm ② — Following-Based Feed



Following Feed Logic

Posts from followed accounts are shown as a blend of latest and oldest entries.

Random Selection between ① & ②

On each session load, the system randomly picks between Algorithm ① (Interest) and Algorithm ② (Following) to prevent feed fatigue and boost discovery.

2.3 Algorithm Rules & Constraints

Trending Criteria

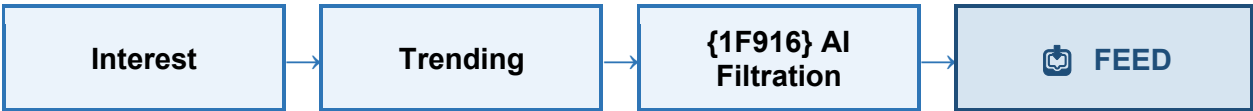
System Constraints

• Most Liked • Most Shared • Most Saved • Most Comments	• Scroll Limit: up to 20 items per session • New User: defaults to both algorithms immediately after first login or account creation
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3. Reels/Vines & Posts Algorithm

The same core algorithm applies to both video and photo posts. The pipeline is: Interest → Trending → Quality → Feed. Location is used as a context signal throughout.

3.1 Main Algorithm Pipeline



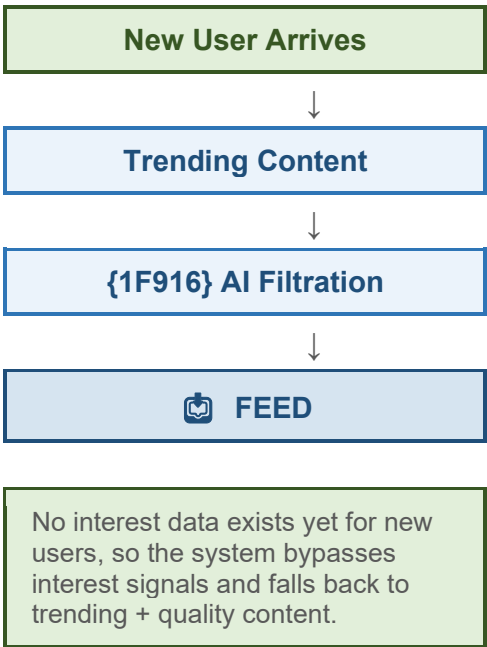
Same algorithm used for both Photos and Vines • Based on Location and Interest

3.2 Trending Signal Ranking

Rank	Signal	Priority Level
1	Watch Time	★ HIGHEST PRIORITY
2	Shares / Saves	High
3	Likes	Medium
4	Comments	Medium

3.3 User State Flowcharts

① New User Flow



② Returning User Flow

